

SEMIALGEBRAIC SETS

Introduction

In this chapter we prove that the sets definable with parameters in the field $\mathbb{R}$ are just the semialgebraic sets. This characterization of the definable sets is essentially the Tarski-Seidenberg theorem. Here we give Łojasiewicz's proof, which gives rather precise information on semialgebraic sets. The ideas involved will also be useful when we deal with semianalytic sets and their generalizations in the next volume. Łojasiewicz's proof was an eye-opener to me, and motivated the notions of o-minimal structure and cell decomposition. Section 1 contains some lemmas that are also useful elsewhere. In Section 2 we prove the main theorem (2.7), and Section 3 characterizes closed semialgebraic sets.

§1. Thom's lemma and continuity of roots

We start with a simple but useful bound on the zeros of a polynomial with complex coefficients:

(1.1) LEMMA. *Let $\alpha \in \mathbb{C}$ be a zero of the monic polynomial*

$$a_0 + a_1T + \cdots + a_{d-1}T^{d-1} + T^d \in \mathbb{C}[T], \quad d \geq 1.$$

Then $|\alpha| \leq 1 + \max\{|a_i| : i = 0, \dots, d-1\}$.

PROOF. Put $M := \max\{|a_i| : i = 0, \dots, d-1\}$ and suppose $|\alpha| > 1 + M$. Then $|a_0 + a_1\alpha + \cdots + a_{d-1}\alpha^{d-1}| \leq M \cdot (1 + |\alpha| + \cdots + |\alpha|^{d-1}) = M \cdot (|\alpha|^d - 1) / (|\alpha| - 1) < |\alpha|^d$, contradicting $0 = |f(\alpha)| = |a_0 + a_1\alpha + \cdots + a_{d-1}\alpha^{d-1} + \alpha^d|$. $\square$

(1.2) THOM'S LEMMA. *Let $f_1, \dots, f_k \in \mathbb{R}[T]$ be nonzero polynomials such that if $f'_i \neq 0$, then $f'_i \in \{f_1, \dots, f_k\}$. Let $\epsilon : \{1, \dots, k\} \rightarrow \{-1, 0, 1\}$, and put*

$$A_\epsilon := \{t \in \mathbb{R} : \text{sign}(f_i(t)) = \epsilon(i), i = 1, \dots, k\}, \text{ a subset of } \mathbb{R}.$$

Then A_ϵ is empty, a point, or an interval. If $A_\epsilon \neq \emptyset$, then its closure is given by

$$\text{cl}(A_\epsilon) = \{t \in \mathbb{R} : \text{sign}(f_i(t)) \in \{\epsilon(i), 0\}, i = 1, \dots, k\}.$$

If $A_\epsilon = \emptyset$, then $\{t \in \mathbb{R} : \text{sign}(f_i(t)) \in \{\epsilon(i), 0\}, i = 1, \dots, k\}$ is empty or a point.

REMARK. We may call ϵ a **sign condition** for $f_1, \dots, f_k$. The 3^k possible sign conditions ϵ determine 3^k disjoint sets A_ϵ , which together cover the real line $\mathbb{R}$. Of course for many ϵ 's the corresponding A_ϵ may be empty. The second statement of the lemma says that for nonempty A_ϵ its closure can be obtained by relaxing all strict inequalities to weak inequalities.

PROOF OF THOM'S LEMMA. By induction on k . The lemma holds trivially for $k = 0$. Let $f_1, \dots, f_k, f_{k+1} \in \mathbb{R}[T] - \{0\}$ be polynomials such that if $f'_i \neq 0$, then $f'_i \in \{f_1, \dots, f_{k+1}\}$. We may as well assume that $\deg(f_{k+1}) = \max\{\deg(f_i) : 1 \leq i \leq k+1\}$. Let $\epsilon' : \{1, \dots, k+1\} \rightarrow \{-1, 0, 1\}$, and let ϵ be the restriction of ϵ' to $\{1, \dots, k\}$. By the inductive hypothesis A_ϵ is empty, a point, or an interval. If A_ϵ is empty or a point, so is $A_{\epsilon'} := A_\epsilon \cap \{t \in \mathbb{R} : \text{sign}(f_{k+1}(t)) = \epsilon'(k+1)\}$, and the other properties to be checked in this case follow easily from the inductive hypothesis on A_ϵ . Suppose A_ϵ is an interval. Since f'_{k+1} has a constant sign on A_ϵ , the function f_{k+1} is either strictly monotone on A_ϵ , or constant. In both cases it is routine to check that $A_{\epsilon'} = A_\epsilon \cap \{t \in \mathbb{R} : \text{sign}(f_{k+1}(t)) = \epsilon'(k+1)\}$ has the required properties. $\square$

(1.3) LEMMA (CONTINUITY OF ROOTS). Let $f(T) = a_0 + a_1T + \dots + a_dT^d \in \mathbb{C}[T]$ be a polynomial that has no zero on the boundary circle $|z - c| = r$ of a given open disc $|z - c| < r$ in the complex plane ($c \in \mathbb{C}$, $r > 0$). Then there is $\epsilon > 0$ such that if $|a_i - b_i| \leq \epsilon$ for $i = 0, \dots, d$, then $g(T) := b_0 + b_1T + \dots + b_dT^d \in \mathbb{C}[T]$ also has no zero on the circle, and f and g have the same number of zeros in the disc. (Here and in the proof below we count zeros with their multiplicity.)

PROOF. From complex analysis we know that the number of zeros of f in the disc $|z - c| < r$ equals

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz,$$

where the circle C is parametrized counterclockwise as $c + re^{2\pi i\theta}$, $0 \leq \theta \leq 1$. By taking ϵ sufficiently small we can clearly guarantee that g has no zeros on C as well, and then the number of zeros of g in the disc is given by the corresponding integral of $(1/2\pi i)(g'(z)/g(z))dz$ over C . Moreover, for small enough ϵ the difference $|(f'/f) - (g'/g)|$ will be so small everywhere on C that the values of the two integrals over C must differ by less than 1 in absolute value. Since these values are integers, they must be equal. $\square$

(1.4) EXERCISES.

1. Let $f = f(X_1, \dots, X_m) \in F[X_1, \dots, X_m]$ be a polynomial with coefficients in the field F , and let $d_1, \dots, d_m \in \mathbb{N}$ be such that $\deg_{X_i} f \leq d_i$ for $i = 1, \dots, m$. Show that if f vanishes identically on a cartesian product $A_1 \times \dots \times A_m$ with $|A_1| > d_1, \dots, |A_m| > d_m$ (all $A_i \subseteq F$), then $f = 0$.

2. Let F be an ordered field and $f \in F[X_1, \dots, X_m]$, $f \neq 0$. Show that the zero set $Z(f) := \{a \in F^m : f(a) = 0\}$ is a closed subset of F^m with empty interior.

§2. Semialgebraic cell decomposition

(2.1) We now come to the setting for Lojasiewicz's theorem:

X is a nonempty topological space, $\mathbf{E}$ a ring of continuous real-valued functions $f: X \rightarrow \mathbb{R}$, the ring operations being pointwise addition and multiplication, with multiplicative identity the function on X that takes the constant value 1.

Call a set $A \subseteq X$ an $\mathbf{E}$ -set if A is a finite union of sets of the form

$$\{x \in X : f(x) = 0, g_1(x) > 0, \dots, g_k(x) > 0\}, \text{ with } f, g_1, \dots, g_k \in \mathbf{E}.$$

Note that the $\mathbf{E}$ -sets form a boolean algebra of subsets of X .

(2.2) EXAMPLE. If $X = \mathbb{R}^m$, and $\mathbf{E} = \mathbb{R}[x_1, \dots, x_m]$ is the ring of real polynomial functions on X , then the $\mathbf{E}$ -sets are exactly the semialgebraic subsets of $\mathbb{R}^m$.

(2.3) Note that the pair $(X \times \mathbb{R}, \mathbf{E}[T])$ also satisfies the conditions imposed on $(X, \mathbf{E})$, where $X \times \mathbb{R}$ is given the product topology, and each polynomial

$$f(T) = f_0 + f_1 T + \dots + f_d T^d \in \mathbf{E}[T]$$

is interpreted as the (continuous) function $(x, t) \mapsto f_d(x)t^d + \dots + f_0(x): X \times \mathbb{R} \rightarrow \mathbb{R}$.

NOTATION. For $f(T)$ as above and $x \in X$, we write $f(x, T)$ for the polynomial

$$f(x, T) = f_0(x) + f_1(x)T + \dots + f_d(x)T^d \in \mathbb{R}[T].$$

(2.4) LEMMA. Suppose X is connected. Let $f = f_0 + f_1 T + \dots + f_d T^d \in \mathbf{E}[T]$, and suppose $e \leq d$ is such that the polynomial $f(x, T) \in \mathbb{R}[T]$ has exactly e distinct complex zeros, for each $x \in X$. Then the number of distinct real zeros of $f(x, T)$ is also constant as x ranges over X . Writing $\zeta_1(x) < \dots < \zeta_r(x)$ for these real zeros, the functions $\zeta_j: X \rightarrow \mathbb{R}$ are continuous.

REMARK. Zeros are counted here *without* multiplicity.

PROOF. Let $x_0 \in X$ and let $z_1, \dots, z_e$ be the distinct complex zeros of $f(x_0, T)$. Take closed balls B_i centered at z_i in the complex plane $\mathbb{C}$, such that $B_i \cap B_j = \emptyset$ for $i \neq j$, and $B_i \cap \mathbb{R} = \emptyset$ if $z_i \notin \mathbb{R}$. By "continuity of roots" there is a neighborhood U of x_0 in X such that for each $x \in U$ the ball B_i contains at least one zero $\zeta_i(x)$ of $f(x, T)$. Note that then $\zeta_i(x)$ is in fact the only zero of $f(x, T)$ in B_i . The graph

of ζ_i on U is $\{(x, t) \in U \times B_i : f(x, t) = 0\}$, hence this graph is closed in $U \times B_i$; in combination with the compactness of B_i this implies that ζ_i is continuous on U .

Since the coefficients of $f(x, T)$ are real, the set $\{\zeta_1(x), \dots, \zeta_e(x)\}$ is closed under complex conjugation. Hence, if $\zeta_i(x_0) = z_i \in \mathbb{R}$, then $\zeta_i(x) \in \mathbb{R}$ for all $x \in U$. This shows that the number of real zeros is locally constant, and thus constant, since X is connected. The real $\zeta_i(x)$'s must also keep their order on the real line as x runs through U . $\square$

(2.5) The lemma below is a special case of Chevalley's constructibility theorem, also known as quantifier elimination for algebraically closed fields, to be proved in the next volume. We give here Łojasiewicz's direct proof of this special case.

(2.6) LEMMA. Let $A = (A_0, \dots, A_d)$ be a tuple of distinct variables and let

$$f(A, T) = A_0 + A_1 T + \dots + A_d T^d \in \mathbb{Z}[A, T]$$

be the general polynomial of degree d . Let $e \in \{0, \dots, d\} \cup \{\infty\}$. Then the set

$$\{a = (a_0, \dots, a_d) \in \mathbb{C}^{d+1} : f(a, T) \text{ has exactly } e \text{ distinct complex zeros}\}$$

is a finite union of sets of the form

$$\{a \in \mathbb{C}^{d+1} : p_1(a) = \dots = p_k(a) = 0, q(a) \neq 0\}, \text{ where } p_i(A), q(A) \in \mathbb{Z}[A].$$

Note that for $e = 0$ the set is $\{(a_0, 0, \dots, 0) : a_0 \neq 0\}$; for $e = \infty$ the set is just the one-point set $\{(0, 0, \dots, 0)\}$. This lemma implies in particular that for $f(T) = f_0 + f_1 T + \dots + f_d T^d \in \mathbb{E}[T]$ the set $\{x \in X : f(x, T) \text{ has exactly } e \text{ complex zeros}\}$ is an $\mathbf{E}$ -set.

PROOF OF (2.6). Let $d > 0$, and $(a_0, \dots, a_d) \in \mathbb{C}^{d+1}$ with $a_d \neq 0$. Put $g := \text{degree of } \gcd(f(a, T), (\partial f / \partial T)(a, T))$ in $\mathbb{C}[T]$; then the lcm of these two polynomials has degree $2d - g - 1$, and the number of distinct complex zeros of $f(a, T)$ is $d - g$. Let $0 < k < d$. Hence the condition

$$(*) \quad \begin{cases} f(a, T) \cdot q(x, T) = (\partial f / \partial T)(a, T) \cdot r(x, T) \text{ for some nonzero} \\ x = (x_0, \dots, x_{2k}) \in \mathbb{C}^{2k+1}, \text{ where} \\ q(x, T) = x_0 + x_1 T + \dots + x_{k-1} T^{k-1} \text{ and} \\ r(x, T) = x_k + x_{k+1} T + \dots + x_{2k} T^k \end{cases}$$

is equivalent to $2d - g - 1 \leq d + k - 1$, that is, to $d - g \leq k$, that is, to the condition that $f(a, T)$ has at most k distinct zeros. Clearly we have

$$\begin{aligned} f(a, T) \cdot q(x, T) - (\partial f / \partial T)(a, T) \cdot r(x, T) \\ = \beta_0(a, x) + \beta_1(a, x)T + \dots + \beta_{d+k-1}(a, x)T^{d+k-1} \end{aligned}$$

for certain bilinear functions $\beta_0, \dots, \beta_{d+k-1} : \mathbb{C}^{d+1} \times \mathbb{C}^{2k+1} \rightarrow \mathbb{C}$.

Hence $(*)$ is equivalent to the condition that $\beta_i(a, x) = 0$ for all i , and some nonzero $x \in \mathbb{C}^{2k+1}$, that is, to the condition that the linear map

$$x \mapsto (\beta_0(a, x), \dots, \beta_{d+k-1}(a, x)) : \mathbb{C}^{2k+1} \rightarrow \mathbb{C}^{d+k}$$

has nontrivial kernel. This last condition is equivalent to the vanishing of all $(2k+1) \times (2k+1)$ minors of the matrix of this linear map. This exhibits the set

$$\{a \in \mathbb{C}^{d+1} : a_d \neq 0 \text{ and } f(a, T) \text{ has at most } k \text{ distinct complex zeros}\}$$

as the intersection of the set $\{a \in \mathbb{C}^{d+1} : a_d \neq 0\}$ with the 'zero set of certain polynomials in $\mathbf{Z}[A]$. The desired result now follows easily. $\square$

(2.7) THEOREM. *Let $f_1(T), \dots, f_M(T) \in \mathbf{E}[T]$. Then the list $f_1, \dots, f_M$ can be augmented to a list $f_1, \dots, f_N$ in $\mathbf{E}[T]$ ($M \leq N$), and X can be partitioned into finitely many $\mathbf{E}$ -sets X_i ($1 \leq i \leq k$) such that for each connected component C of each X_i there are continuous real-valued functions $\xi_{C,1} < \dots < \xi_{C,m(C)}$ on C with the following two properties:*

- (1) *each function f_n ($1 \leq n \leq N$) has constant sign ($-1, 0$, or 1) on each of the graphs $\Gamma(\xi_{C,j})$ ($1 \leq j \leq \mu(C)$) and on each of the sets $(\xi_{C,j}, \xi_{C,j+1})$ ($0 \leq j \leq \mu(C)$), where $\xi_{C,0} := -\infty$ and $\xi_{C,m(C)+1} := +\infty$ are constant functions on C by convention;*
- (2) *each of the sets $\Gamma(\xi_{C,j})$ and $(\xi_{C,j}, \xi_{C,j+1})$ from (1) is of the form*

$$\{(x, t) \in C \times \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$$

for a suitable sign condition $\epsilon : \{1, \dots, N\} \rightarrow \{-1, 0, 1\}$.

PROOF. Take $d \in \mathbb{N}$ such that each of the polynomials $f_m(T)$ is of the form

$$f_m(T) = f_{m0} + f_{m1}T + \dots + f_{md}T^d, \text{ with } f_{mr} \in \mathbf{E} \text{ for } 0 \leq r \leq d.$$

Let Δ range over the subsets of $\{1, \dots, M\} \times \{0, \dots, d\}$, and put

$$f_\Delta(T) := \prod_{(m,r) \in \Delta} (\partial^r f_m / \partial T^r) \in \mathbf{E}[T], \text{ so } \deg_T(f_\Delta) \leq Md^2.$$

We let e range over $\{0, \dots, Md^2, \infty\}$, and put $A_{\Delta_e} := \{x \in X : f_\Delta(x, T) \text{ has exactly } e \text{ complex zeros}\}$. From the remark following lemma (2.6) we obtain that A_{Δ_e} is an $\mathbf{E}$ -set. Clearly, for a given Δ the sets A_{Δ_e} are pairwise disjoint as e varies, and cover X . Because the $\mathbf{E}$ -sets form a boolean algebra we can find a partition $X = X_1 \cup \dots \cup X_k$ of X into $\mathbf{E}$ -sets X_i such that each set A_{Δ_e} is a union of X_i 's. Augment $f_1, \dots, f_M$ to $f_1, \dots, f_N$ such that

$$\{f_1, \dots, f_N\} = \{\partial^r f_m / \partial T^r : 1 \leq m \leq M, 0 \leq r \leq d\}.$$

We shall prove that this sequence $f_1, \dots, f_N$ and these X_i 's satisfy the conclusion of our theorem.

For each connected component C of an X_i , let $\Delta(C)$ be the set

$$\left\{ (m, r) \in \{1, \dots, M\} \times \{0, \dots, d\} : \frac{\partial^r f_m}{\partial T^r} \text{ does not vanish identically on } C \times \mathbb{R} \right\}.$$

Fix such a component C . Given any Δ , there exists e such that $C \subseteq A_{\Delta e}$, in particular C is contained in $A_{\Delta(C)e}$ for some e . Then $f_{\Delta(C)}(x, T)$ has exactly e complex zeros, for each $x \in C$. We claim e is finite. To see this, consider a factor $g(T) = \partial^r f_m / \partial T^r$ of $f_{\Delta(C)}$, $(m, r) \in \Delta(C)$. The set C is contained in $A_{\{(m, r)\}e'}$ for some $e' \in \{0, \dots, d, \infty\}$, so the number e' of complex zeros of $g(x, T)$ is independent of $x \in C$. Because $(m, r) \in \Delta(C)$, this number e' is finite, and hence e is finite. Lemma (2.4) now implies there are continuous real-valued functions $\xi_{C,1} < \dots < \xi_{C,\mu(C)}$ on C such that

$$\{(x, t) \in C \times \mathbb{R} : f_{\Delta(C)}(x, t) = 0\} = \Gamma(\xi_{C,1}) \cup \dots \cup \Gamma(\xi_{C,\mu(C)}).$$

We shall prove that these functions $\xi_{C,j}$ satisfy the conclusions of the theorem.

CLAIM 1. Each function $\partial^r f_m / \partial T^r$ ($1 \leq m \leq M$, $0 \leq r \leq d$) has constant sign on each set $\Gamma(\xi_{C,j})$ ($1 \leq j \leq \mu(C)$) and $(\xi_{C,j}, \xi_{C,j+1})$ ($0 \leq j \leq \mu(C)$), where $\xi_{C,0} = -\infty$ and $\xi_{C,\mu(C)+1} = +\infty$.

Note that this claim gives us conclusion (1) of the theorem, since each f_n is one of the $\partial^r f_m / \partial T^r$'s. To prove the claim, note first that $\partial^r f_m / \partial T^r$ vanishes identically on the indicated sets if $(m, r) \notin \Delta(C)$. Let $(m, r) \in \Delta(C)$. Then, with $g(T) := \partial^r f_m / \partial T^r$, the number of complex zeros of $g(x, T)$ is independent of $x \in C$, as we already saw. Hence by lemma (2.4) the zero set of g on $C \times \mathbb{R}$ is a finite disjoint union of graphs of continuous real-valued functions on C . Since $f_{\Delta(C)}$ contains g as a factor, it is clear that these functions must be among the $\xi_{C,j}$'s. (Use that C is connected.) This proves claim 1.

Now let B be one of the sets in claim 1. Put $\epsilon(m, r) := \text{sign}(\partial^r f_m / \partial T^r)$ on B , and put

$$B' := \left\{ (x, t) \in C \times \mathbb{R} : \text{sign} \left(\frac{\partial^r f_m}{\partial T^r} \right) (x, t) = \epsilon(m, r) \text{ for } 1 \leq m \leq M, 0 \leq r \leq d \right\}.$$

Clearly $B \subseteq B'$.

CLAIM 2. $B = B'$.

Suppose not. Take $(x, t') \in B' - B$ and $(x, t) \in B$. Say $t < t'$. Thom's lemma implies that $\{s \in \mathbb{R} : (x, s) \in B'\}$ is connected, so $\{x\} \times [t, t'] \subseteq B'$. Note that $f_{\Delta(C)}$ must change sign on $\{x\} \times [t, t']$ since $(x, t) \in B$ and $(x, t') \notin B$. But $f_{\Delta(C)}$ is a product of

partials $\partial^r f_m / \partial T^r$, so $f_{\Delta(C)}$ cannot change sign on B' , contradiction. This proves claim 2, and thus conclusion (2) of the theorem. $\square$

(2.8) Let us say that the pair $(X, \mathbf{E})$ has the **Łojasiewicz property** if each $\mathbf{E}$ -set has only finitely many connected components, and each component is also an $\mathbf{E}$ -set.

(2.9) COROLLARY. *If $(X, \mathbf{E})$ has the Łojasiewicz property, then $(X \times \mathbb{R}, \mathbf{E}[T])$ also has the Łojasiewicz property, and the image of any $\mathbf{E}[T]$ -set $S \subseteq X \times \mathbb{R}$ under the projection map $X \times \mathbb{R} \rightarrow X$ is an $\mathbf{E}$ -set.*

PROOF. Write an $\mathbf{E}[T]$ -set S as a finite union of sets of the form

$$\{(x, t) \in X \times \mathbb{R} : f(x, t) = 0, g_1(x, t) > 0, \dots, g_k(x, t) > 0\}$$

with $f, g_1, \dots, g_k \in \mathbf{E}[T]$. Let $f_1, \dots, f_M$ be a list of all the polynomials in $\mathbf{E}[T]$ that are involved in such a definition of S . Now apply the theorem to this list. $\square$

(2.10) The $\mathbb{R}[T_1, \dots, T_m]$ -sets in the space $\mathbb{R}^m$ are exactly the semialgebraic subsets of $\mathbb{R}^m$. For $m = 0$ the space $\mathbb{R}^m$ has only one point, so the pair $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$ obviously has the Łojasiewicz property for $m = 0$; hence by induction on m , using (2.9), it follows that $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$ has the Łojasiewicz property for each m , and that the image of a semialgebraic subset of $\mathbb{R}^{m+1}$ under the projection map $\mathbb{R}^{m+1} \rightarrow \mathbb{R}^m$ is a semialgebraic subset of $\mathbb{R}^m$. (Tarski-Seidenberg property)

We can now draw the following conclusion about the model-theoretic structure $(\mathbb{R}, <, 0, 1, +, -, \cdot)$, which we will call “the ordered field of real numbers”.

(2.11) COROLLARY. *The sets $S \subseteq \mathbb{R}^m$ ($m = 0, 1, 2, \dots$) that are definable using constants in the ordered field of real numbers are exactly the semialgebraic sets. The ordered field of real numbers is o-minimal.*

PROOF. By (2.10) the semialgebraic sets form a structure on $(\mathbb{R}, <)$. Clearly the primitives of the ordered field of reals are semialgebraic. By Chapter 1, (5.9), exercise 3, the semialgebraic sets are definable using constants in the ordered field of real numbers. Finally, the semialgebraic subsets of $\mathbb{R}$ are clearly the finite unions of intervals and points. The desired result follows. $\square$

This classical example should be kept in mind throughout this book. In the next volume we extend this result by showing that all real closed fields are o-minimal, in a uniform way, and indicate how this fact throws light on the classical real case as well. There we also treat the remarkable recent theorem by Wilkie that the ordered *exponential* field of reals is o-minimal, and its aftermath.

§3. Thom's lemma with parameters

This last section is mainly a preparation for our later work on semianalytic sets in a later volume. We return to the set-up where we have a pair $(X, \mathbf{E})$ as before: X a nonempty topological space, $\mathbf{E}$ a ring of real-valued continuous functions on X .

(3.1) DEFINITION. Let $f_1(T), \dots, f_M(T) \in \mathbf{E}[T]$, and let C be a connected subset of X . A **decomposition of $f_1, \dots, f_M$ above C** consists of continuous real-valued functions $\xi_1 < \dots < \xi_r$ on C such that

- (1) each function f_m has constant sign on each set $\Gamma(\xi_i)$ ($1 \leq i \leq r$) and on each set (ξ_i, ξ_{i+1}) ($0 \leq i \leq r$, where $\xi_0 := -\infty$, $\xi_{r+1} := +\infty$);
- (2) each set $\Gamma(\xi_i)$ is contained in the zero set of some function f_m .

Note: if (1) holds, then (2) can be satisfied by removing the ξ_i 's not satisfying (2). Theorem (2.7) implies that if $(X, \mathbf{E})$ has the Łojasiewicz property, then for every sequence $f_1, \dots, f_M$ in $\mathbf{E}[T]$ there is a partition of X into finitely many connected $\mathbf{E}$ -sets, above each of which $f_1, \dots, f_M$ has a decomposition.

Let us say that a polynomial $f(T) \in \mathbf{E}[T]$ **has zero-free leading coefficient** if $f(T) = c_d T^d + c_{d-1} T^{d-1} + \dots + c_0$ with $c_i \in \mathbf{E}$, such that c_d has no zero on X .

By (1.1) the real zeros of a polynomial $a_d T^d + a_{d-1} T^{d-1} + \dots + a_0 \in \mathbb{R}[T]$ with $a_d \neq 0$ are in absolute value $\leq 1 + \max\{|a_i/a_d| : 0 \leq i \leq d-1\}$.

Hence if $f(T) \in \mathbf{E}[T]$ has zero-free leading coefficient and a point $y \in X$ is given, then the real zeros of $f(x, T)$ are bounded in absolute value by some constant, for all x in a suitably small neighborhood of y .

This fact is used in the proof of the next lemma, which constitutes what one might call "Thom's lemma with parameters".

(3.2) LEMMA. Suppose the polynomials $f_1, \dots, f_N \in \mathbf{E}[T]$ have zero-free leading coefficients, and each nonzero partial $\partial f_n / \partial T$ also belongs to $\{f_1, \dots, f_N\}$. Let $\xi_1 < \dots < \xi_r$ be a decomposition of $f_1, \dots, f_N$ above a connected set $C \subseteq X$. Fix $i \in \{1, \dots, r\}$, let $\xi = \xi_i$, put $\epsilon(n) = \text{sign}(f_n | \Gamma(\xi))$ for $1 \leq n \leq N$. Then

- (a) $\Gamma(\xi) = \{(x, t) \in C \times \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$,
- (b) $\xi : C \rightarrow \mathbb{R}$ extends uniquely to a continuous function $\eta : \text{cl}(C) \rightarrow \mathbb{R}$, and $\text{cl}(\Gamma(\xi)) = \Gamma(\eta) = \{(y, t) \in \text{cl}(C) \times \mathbb{R} : \text{sign}(f_n(y, t)) \in \{\epsilon(n), 0\} \text{ for } n = 1, \dots, N\}$.

PROOF. For fixed $x \in C$ the set

$$A_{x, \epsilon} := \{t \in \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$$

contains $\{\xi(x)\}$ and is finite since some f_m vanishes on $\Gamma(\xi)$ and $f_m(x, T) \neq 0$. The polynomials $f_1(x, T), \dots, f_N(x, T)$ satisfy the hypothesis of Thom's lemma, so $A_{x, \epsilon}$ is connected. Hence $A_{x, \epsilon} = \{\xi(x)\}$. This proves (a).

For (b), take m such that $f_m(x, \xi(x)) = 0$ for all $x \in C$. Let $y \in \text{cl}(C)$. Since $f_m(T)$ has zero-free leading coefficient, there is a neighborhood U of y such that ξ is bounded on $U \cap C$. Hence the limits

$$l := \liminf_{x \rightarrow y} \xi(x) \\ \text{and } L := \limsup_{x \rightarrow y} \xi(x)$$

are real numbers (not $\pm\infty$). Now $\text{sign}(f_n(x, \xi(x))) = \epsilon(n)$ for all n , hence $\text{sign}(f_n(y, l)) \in \{\epsilon(n), 0\}$, and $\text{sign}(f_n(y, L)) \in \{\epsilon(n), 0\}$ for all n . By Thom's lemma the set $\{t \in \mathbb{R} : \text{sign}(f_n(y, t)) \in \{\epsilon(n), 0\} \text{ for } n = 1, \dots, N\}$ is connected, so this set contains $[l, L]$. With m as above we have $f_m(y, t) = 0$ for all $t \in [l, L]$, and because $f_m(y, T) \neq 0$, this implies $l = L$. Since y was an arbitrary point of $\text{cl}(C)$ we have shown that $\eta(y) := \lim_{x \rightarrow y} \xi(x)$ defines a function $\eta : \text{cl}(C) \rightarrow \mathbb{R}$. Continuity of ξ implies continuity of η , so $\Gamma(\eta)$ is closed, hence $\text{cl}(\Gamma(\xi)) = \Gamma(\eta)$.

The argument with Thom's lemma that we just gave also shows

$$\Gamma(\eta) = \{(y, t) \in \text{cl}(C) \times \mathbb{R} : \text{sign}(f_n(y, t)) \in \{\epsilon(n), 0\} \text{ for } n = 1, \dots, N\}.$$

This finishes the proof of (b). $\square$

(3.3) LEMMA. *With the same assumptions as in the previous lemma, set $\xi_0 := -\infty$, $\xi_{r+1} := +\infty$. Fix an $i \in \{0, \dots, r\}$ and put $\epsilon(n) = \text{sign}(f_n((\xi_i, \xi_{i+1})))$. Then we have*

- (a) $(\xi_i, \xi_{i+1}) = \{(x, t) \in C \times \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$,
- (b) $\text{cl}(\xi_i, \xi_{i+1}) = \{(y, t) \in \text{cl}(C) \times \mathbb{R} : \text{sign}(f_n(y, t)) \in \{\epsilon(n), 0\} \text{ for } n = 1, \dots, N\}$.

We leave this as an exercise.

(3.4) FURTHER TERMINOLOGY. Let F be a finite set of continuous real-valued functions on a topological space X . We say that a set $S \subseteq X$ is **described by F** if S is a finite union of sets of the form

$$\epsilon(F) := \{x \in X : \text{sign}(f(x)) = \epsilon(f) \text{ for all } f \in F\}, \text{ with } \epsilon : F \rightarrow \{-1, 0, 1\}.$$

Since F is a finite collection of functions, the sets described by F form a finite boolean algebra of subsets of X .

(3.5) NOETHER NORMALIZATION. Let $f(X_1, \dots, X_n)$ be a nonzero polynomial of degree d over a field K of characteristic 0, say $f = \sum_{|i|=d} c_i X^i + \text{terms of lower degree}$, $n > 0$. Here $i = (i_1, \dots, i_n) \in \mathbb{N}^n$, $|i| = i_1 + \dots + i_n$ and $X^i = X_1^{i_1} \dots X_n^{i_n}$, $c_i \in K$. Let $(\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$. We are going to make the (invertible) substitution

$$\begin{aligned} X_1 &\rightarrow X_1 + \lambda_1 X_n, \\ &\vdots \\ X_{n-1} &\rightarrow X_{n-1} + \lambda_{n-1} X_n, \\ X_n &\rightarrow X_n. \end{aligned}$$

Note that, with $\lambda^i = \lambda_1^{i_1} \cdots \lambda_{n-1}^{i_{n-1}}$, this substitution transforms f into

$$\begin{aligned} & f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n) \\ &= \left(\sum_{|i|=d} c_i \lambda^i \right) X_n^d + \text{terms of degree } < d \text{ in } X_n. \end{aligned}$$

Since the polynomial $\sum_{|i|=d} c_i Y_1^{i_1} \cdots Y_{n-1}^{i_{n-1}} \in K[Y_1, \dots, Y_{n-1}]$ is nonzero, it follows that for “almost all” $\lambda = (\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$, namely for all those λ that are not zeros of this polynomial, the transformed polynomial

$$f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n)$$

is of (total) degree d , and has a term of the form cX_n^d with $c \in K - \{0\}$.

(3.6) COROLLARY. *A closed semialgebraic subset of $\mathbb{R}^n$ is a finite union of sets of the form $\{x \in \mathbb{R}^n : f_1(x) \geq 0, \dots, f_k(x) \geq 0\}$ with $f_1, \dots, f_k \in \mathbb{R}[X_1, \dots, X_n]$.*

PROOF. By induction on n . The case $n = 0$ is trivial; assume the desired result holds for a certain n , let $S \subseteq \mathbb{R}^{n+1}$ be a closed semialgebraic set, and let F be a finite set of nonzero polynomials in $\mathbb{R}[X_1, \dots, X_n, T]$ describing S . By a linear transformation of variables as in (3.5) we may as well assume that all $f \in F$ are monic in T up to a constant factor, and hence, by enlarging F , we may also assume that each nonzero partial $\partial^r f / \partial T^r$ of each $f \in F$ belongs to F . By theorem (2.7) and by (2.10) there is a partition $\mathcal{P}$ of $\mathbb{R}^n$ into finitely many connected semialgebraic sets C such that F has a decomposition over each $C \in \mathcal{P}$. Now apply the inductive assumption to the closed semialgebraic sets $\text{cl}(C)$, and apply lemmas (3.2) and (3.3) to get the desired result. $\square$

(3.7) EXERCISES.

1. Let $Q(X, Y) \in \mathbb{R}[X, Y]$ be a nonzero polynomial in two variables. Show that there are $d \in \mathbb{N}$ and $M > 0$ such that if $(x, y) \in \mathbb{R}^2$, $x > M$ and $Q(x, y) = 0$, then $|y| \leq x^d$.

In the next two problems we say that a map $g: A \rightarrow \mathbb{R}^n$ with $A \subseteq \mathbb{R}^m$ is **semialgebraic** if its graph $\Gamma(g) \subseteq \mathbb{R}^{m+n}$ is semialgebraic.

2. Show that if $g: \mathbb{R} \rightarrow \mathbb{R}$ is semialgebraic, then there are $d \in \mathbb{N}$ and $M > 0$ such that $|g(x)| \leq x^d$ for all $x > M$.

3. Suppose a continuous function $g: \mathbb{R}^m \rightarrow \mathbb{R}$ satisfies $Q(x, g(x)) = 0$ for all $x \in \mathbb{R}^m$ and some nonzero polynomial $Q(X, T) \in \mathbb{R}[X, T]$, $X = (X_1, \dots, X_m)$. Show that then g is semialgebraic.

Notes and comments

Classical sources for much of the material in this chapter are Koopman and Brown [36], where analytic varieties are locally decomposed into analytic cells, Whitney [63], which stratifies real algebraic sets into finitely many connected semialgebraic manifolds, and Łojasiewicz [40], where the Tarski-Seidenberg theorem is obtained by cell decomposition. See also the books by Benedetti and Risler [2] and by Bochnak, Coste and Roy [4].

The proof of (3.6) given here follows the proof of a corresponding local result in Łojasiewicz [40]. Other proofs are by Bochnak and Efroymsen [5], Delzell [14], Coste and Roy, see [4]; for a short model-theoretic proof, see [17].

The result of exercise 2 in (3.7) means that the ordered field of real numbers is **polynomially bounded**. This property is shared by other o-minimal expansions of this structure, but of course not by $(\mathbb{R}, <, 0, 1, +, -, \cdot, \exp)$. Miller [43] discovered the surprising fact that every o-minimal expansion of the ordered field of reals is either polynomially bounded, or defines the exponential function. Polynomial boundedness implies the various Łojasiewicz inequalities, see [23].

